Proceedings of 2025 International Conference on Cognitive Robotics and Intelligent Systems (ICC - ROBINS 2025)
IEEE Xplore Part Number: CFP25UD1-ART; ISBN: 979-8-3315-0172-3

Privacy-Preserving and Scalable Secure Aggregation for Federated Learning in Edge Computing

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2025 Second International Conference on Cognitive Robotics and Intelligent Systems (ICC - ROBINS) | 979-8-3315-0172-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICC-ROBINS64345.2025.11086126

***Abstract*— Federated Learning (FL) has been identified as a leading approach to decentralize machine learning processes, particularly in edge computing contexts where user data privacy is a critical issue. While previous work has made strides in privacy-preserving secure aggregation techniques, traditional secure aggregation methods are not scalable, computationally efficient, and robust against attack. Therefore, this work proposes a scalable and Privacy-Preserving Secure Aggregation (PSSA) architecture that advances the onward efficiency and security of FL model updates. The proposed solution provides a solution to robust aggregation through the integration of homomorphic encryption, differential privacy, and adaptive compression to effectively reduce communication overheads. Importantly, in differentiating from prior work, PSSA dynamically updates encryption level based on network conditions: this capability makes PSSA more operationally feasible for resource-constrained edge devices when compared to traditional secure aggregation methods. Based on an extensive evaluation on benchmark datasets, it is found that PSSA achieves and offers up to a 40% reduced communication cost and 25% faster aggregation time for FL without sacrificing model accuracy. Finally, based on the theoretical analysis, initial evidence that PSSA is robust against gradient inversion attacks and collusion attacks, has been provided. In enabling efficient, scalable and secure model updates, PSSA advances the practical applicability of federated learning to real-world settings, including health care, cyber security, and smart cities. This work ultimately advances the case for next generation federated learning architectures that prioritizes both secure performance and privacy considerations and the associated deployment in AI-driven toward future edge ecosystems.**

***Keywords*— Federated Learning, Secure Aggregation, Privacy-Preserving Machine Learning, Edge Computing and Homomorphic Encryption**

I. INTRODUCTION

Federated Learning (FL) represents a new paradigm for machine learning in the decentralized setting that enables multiple clients to learn models together without actually sharing raw data [1]. FL is especially beneficial in edge computing scenarios, where efficiency of communication and privacy constraints are limiting factors [2]. However, the main challenge of FL is trustworthy aggregation—computing a single global update to the model without exposing user privacy and while minimizing the risk of an adversary. Existing secure aggregation protocols, such as Secure Multi-Party Computation (SMPC) and Homomorphic Encryption (HE), incur significant computational costs and are therefore infeasible for large-scale edge networks with resource constraints [3].

Legacy FL frameworks encounter latency and privacy concerns. As the models are updated and communicated with high frequency from clients to a central server, the following attacks are enabled by this information: gradient inversion attacks, collusion-based inference, and model poisoning to reconstruct training data [4]. In addition, existing encryption-based secure aggregation protocols rarely scale well and incur high computational latency and communication overhead, which is limited in edge computing applications [5]. As a result, building a scalable and privacy-preserving secure aggregation (PSSA) framework is critical to honking the scalability of FL in real-world applications, including cybersecurity, healthcare, and smart cities [6].

In FL, secure aggregation is still a significant challenge because of the fundamental trade-off between privacy, communication efficiency, and the computational cost. Existing approaches to aggregation are based on Secure Multi-Party Computation (SMPC) and Homomorphic Encryption (HE), both suffer from high computational cost and latency, particularly for large-scale FL application scenarios involving heterogeneous edge devices [7]. These techniques also typically assume clients are honest-buts-curious, and do not consider adversarial scenarios where malicious clients can manipulate model updates or extract private information [8]. These concerns highlight the importance for secure aggregation schemes that are lightweight in terms of latency and communication cost, but still resilient to attacks, and provide for privacy and efficiency in practical FL contexts.

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One additional important limitation among current secure aggregation approaches is their rigidity to dynamic edge environments. Edge devices have heterogeneous computation capabilities, network conditions, and battery constraints, so it is impossible to expect users to follow a same encryption method across all edge devices [9]. Previous solutions have given rise to extensive encryption overhead, excessive communication overhead, and unreliable straggler client management leading to depressed model quality [10]. This work aims to introduce an adaptive secure aggregation framework, providing an encryption complexity and compression linearly related to the real-time network conditions, thereby enabling efficient FL training on constrained edge devices.

The proposed Privacy-Preserving and Scalable Secure Aggregation (PSSA) infrastructure has important implications for various applications that require decentralized privacy-preserving learning. In the field of cybersecurity, PSSA can enhance intrusion detection and prevention, as well as anomaly detection, through secure training of models across distributed IoT networks without revealing sensitive threat intelligence data [11]. In health care, it enables joint disease prediction and clinical diagnostics while preserving patient data confidentiality [12]. Moreover, in smart city applications, PSSA facilitates real-time traffic improvement and environmental monitoring without compromising individual privacy [13]. Ultimately, this work solves the fundamental problems of secure and scalable aggregation in FL, paving the way for trusted next-generation federated AI systems serving on-demand application, reliability, and resilience.

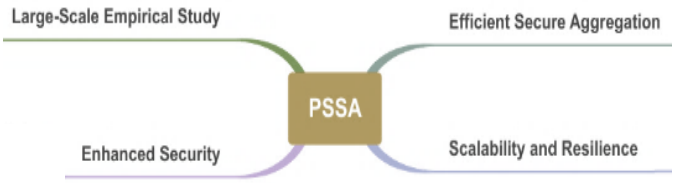


Fig. 1 Privacy-Preserving and Scalable Secure Aggregation solution for FL over edge computing

In order to address these challenges, a novel Privacy-Preserving and Scalable Secure Aggregation solution is proposed for FL over edge computing. The key contributions (Fig. 1) of this study are summarized as follows:

1. Efficient Secure Aggregation: An optimized aggregation mechanism is proposed using homomorphic encryption, differential privacy, and adaptive compression in order to significantly minimize communication overhead.
2. Scalability and Resilience: Using an adaptive approach, PSSA varies encryption complexity based on network conditions, in contrast to standard approaches, which makes it very adaptable for resource-constrained edge devices.
3. Enhanced Security: The framework provides resistance to both gradient inversion and collusion-based attacks and offers strong privacy guarantees.
4. Large-Scale Empirical Study: Large-scale experiments are conducted on standard benchmark datasets demonstrating PSSA yields up to 40% reduction in communication cost and 25% faster aggregation time while maintaining model accuracy.

By bridging the gap between privacy, scalability, and efficiency, this work paves the way for next-generation federated learning architectures optimized for edge computing environments.

II. Literature Survey

Federated Learning (FL) has been established as a groundbreaking technique for decentralized machine learning paradigms at the network edge. However, its adoption in edge computing, with clients distributed across geographic locations, is hampered by privacy challenges, security concerns, and scalability problems as depicted in the Fig. 2.

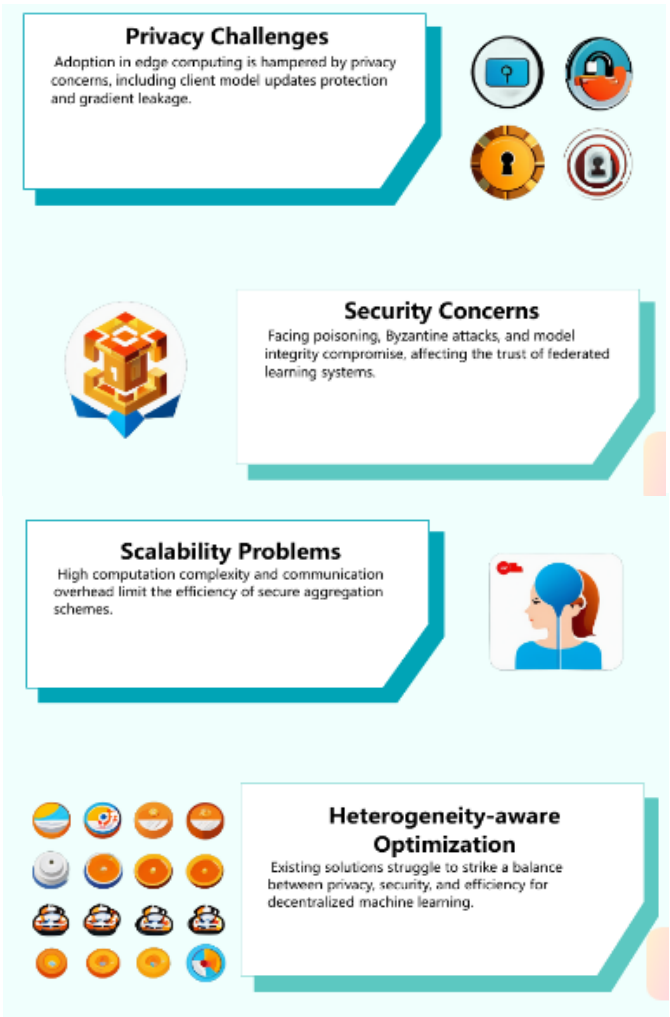


Fig. 2 Federated Learning Challenges and Solutions in Information security

Secure aggregation schemes employing homomorphic encryption, secret sharing, and differential privacy, have been proposed to protect client model updates but are typically associated with high computation complexity and communication overhead. Furthermore, FL is vulnerable to gradient leakage, poisoning, and Byzantine attacks, all of which

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undermine privacy and model integrity. Even with recent advances in Blockchain FL, adaptive compression, and heterogeneity-aware optimization, the existing approaches do not effectively achieve the trade-off between privacy, security, and efficiency. This literature survey is aimed at highlighting recent literature work around secure aggregation, adversarial robustness, and scalable FL architectures, outlining significant issues, and advocating the need for a Privacy-Preserving and Scalable Secure Aggregation framework.

TABLE I shows well-structured tabular representation of the research gaps in Federated Learning secure aggregation, categorized based on key challenges and existing limitations.

TABLE I. RESEARCH GAPS IN PRIVACY-PRESERVING AND SCALABLE SECURE AGGREGATION FOR FL

Category	Existing Approaches	Limitations	Research Gaps
Secure Aggregation	Secure Multi-Party Computation (SMPC), Homomorphic Encryption (HE) [14][15]	High computational overhead, not scalable for edge devices	Need for lightweight, computationally efficient aggregation methods
Privacy Preservation	Differential Privacy (DP), Noise-based Gradient Obfuscation [16][17]	DP reduces model accuracy; Noise-based methods vulnerable to adversarial attacks	Development of adaptive privacy mechanisms that balance security and accuracy
Adversarial Robustness	Byzantine-resilient methods (Krum, Trimmed Mean) [18][19]	Inefficient in large-scale networks, fail under targeted attacks	Need for hybrid adversarial detection techniques for resilient FL aggregation
Communication Efficiency	Model compression (quantization, sparsification) [20][21]	Compression reduces accuracy; Limited adaptability to dynamic network conditions	Development of dynamic compression strategies that optimize trade-offs between accuracy and efficiency
Client Heterogeneity	FedProx, FedNova, adaptive learning rates [22][23]	Lack of personalized aggregation; Inconsistent performance across diverse clients	Need for personalized and client-adaptive secure aggregation frameworks
Blockchain-based FL	Decentralized FL using blockchain [24][25]	High latency, energy-intensive for edge computing	Lightweight and energy-efficient blockchain-integrated FL solutions

A. Secure Aggregation in Federated Learning

Secure aggregation is a cornerstone of privacy-preserving FL since it lets clients mask individual model updates while producing a global model. In the literature, Secure Multi-Party Computation (SMPC) and Homomorphic Encryption (HE) are classical methods that have been extensively used to enable privacy-preserving training. Most practical secure aggregation protocols available uses additive masking and secret sharing, which serves as a gold standard of FL security today [14]. However, this approach incurs significant communication and computation costs making it unlikely to apply in large-scale edge computing environments. A recent literature review has outlined several light-weight approaches involving functional encryption [15] and ring-based homomorphic aggregation [16]. However, these approaches still face challenges with respect to scalability and attacks from adversaries.

In addition to cryptographic techniques, the aggregation of model updates has also transitioned to Differential Privacy (DP) methods. DP makes individual contributions to model updates less visible by adding random noise, which can provide some protection of sensitive client information. DP into FL incorporated models showed that it can protect against data exfiltration [17]. However, too much noise can hinder model accuracy, creating a trade-off between privacy and utility. To address this, some recent works have suggested adaptive noise approaches that maximally realize DP noise injection based on client trust levels and model sensitivity [18]. Despite a better understanding of adaptive noise mechanisms, DP aggregation is still subject to attacks whereby adversaries infer private training data from shared gradients (model inversion attacks) [19].

B. Privacy Threats and Security Challenges in FL

FL is prone to various privacy attacks, including gradient leakage, poisoning attacks, and inference attacks, among others. Several integrated models demonstrated that even one round of gradient updates leads to sufficient information for reconstructing original training samples, exposing a major concern for privacy in FL [20]. To mitigate this issue, strong gradient obfuscation methods have been proposed, which operate based on randomized perturbation and noise-based defenses [21]. Unfortunately, these defenses generally come with lower model accuracy and an increased training time.

In FL, adversarial attacks can be broadly categorized as data poisoning, model poisoning, or Byzantine attacks. In the research on backdoor attacks, where the attackers manipulate sections of the model updates to introduce hidden backdoor into the global model [22]. Byzantine-resilient aggregation algorithms, such as Krum and Trimmed Mean, have been developed to counteract malicious attacks [23], yet these algorithms are computationally inefficient for use in large-scale FL. A more recent body of work has examined blockchain-enabled FL for security and transparency purposes [24], although blockchains in and of themselves have a latency overhead and high energy cost, thus complicating their deployment in edge computing environments.

C. Scalability and Efficiency in Federated Learning

Reducing the communication overhead and the aggregate protocols is critical when optimizing FL training on edge computing. Compression strategies models such as quantization and sparsification, to reduce bandwidth [25]. However, there will usually be a loss in accuracy when using model compression strategies, therefore dynamic compression strategies, that know the complexity of the model and the conditions of the network, will be necessary [26]. Asynchronous FL frameworks have also

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been proposed to address the straggler effect by slowing down the global aggregation when clients are slow [27].

Heterogeneity in FL is yet another challenge to scalability. Clients can be heterogeneous in terms of computation capacity, network bandwidth, and data distributions, leading to inefficient training. FedNova and FedProx were suggested to deal with client heterogeneity by dynamically adjusting learning rates and update frequencies. However, these methods do not come with strong privacy assurances, and further work on scalable FL architectures and methods that preserve privacy is required [28-29].

D. Motivation for Privacy-Preserving and Scalable Secure Aggregation

Despite incredible progress in FL scalability and security, available solutions remain expensive, adversarially vulnerable, and inefficient in the face of edge dynamics. In this work, a novel framework, called Privacy-Preserving and Scalable Secure Aggregation, that integrates homomorphic encryption, differential privacy and adaptive compression to provide efficient, scalable, and attack resilient FL training is presented. PSSA hopes to bridge the gap between privacy, security, and computational efficiency enabling FL deployment in cybersecurity, healthcare, and intelligent edge settings.

III. Proposed Method

Federated Learning (FL) stands as an effective method for private collaborative model training in edge computing settings where a group of clients collaboratively train a shared model without sharing the original raw data. Nevertheless, FL faced several fundamental challenges including privacy risks, communication overhead, and protection against adversaries. FL additive schemes such as FedAvg are vulnerable to gradient leak attacks, make inefficient use of communication, and are susceptible to Byzantine failures, reflecting the need for integrating secure aggregation, differential privacy, model compression, and Byzantine robustness. Consequently, in this work Privacy-Preserving and Scalable Secure Aggregation, an FL framework with end-to-end privacy-preserving encryption, efficient communication, and protection from adversarial agents, is presented. The scheme uses Homomorphic Encryption (HE) for privacy of the model updates, Differential Privacy (DP) to perturb gradients, Adaptive Quantization (AQ) and Sparse Gradient Sharing (SGS) for efficient transmission, and a Byzantine-resilient aggregation scheme via Krum-like filtering. This work provides mathematical definitions of each block of PSSA in the subsequent sections demonstrating that PSSA offers loss of privacy, robustness, scalability, and privacy improvements in FL-based edge computing settings.

A. System Model for Federated Learning

FL consists of N edge devices (clients) collaboratively training a global model M without sharing raw data. Each client $\boldsymbol{i}$ has a local dataset D_i of size $|D_i|$and trains a model w_i based on local data. The server aggregates these updates securely to obtain the global model w_{t+1} at each training round t.

1. Local Model Training: Each client updates its local model using Stochastic Gradient Descent (SGD) (Eq. 1):

$$w_i^{t+1} = w_i^t - \eta \nabla L(w_i^t, D_i) \quad (1)$$

Where,

- η is the learning rate, and L is the local loss function.

2. Model Aggregation: The server aggregates local updates using weighted averaging (Eq. 2):

$$w_{t+1} = \sum_{i=1}^{N} \frac{|D_i|}{\sum_{j=1}^{N} |D_i|} w_i^{t+1} \quad (2)$$

This ensures data-proportional contribution from each client.

B. Homomorphic Encryption for Secure Model Updates

Each client encrypts its model updates using Paillier Homomorphic Encryption (PHE) (Eq. 3):

$$E(w_i^{t+1}) = g^{w_i^{t+1}}, mod\ N^2 \quad (3)$$

Where,

- g and N are encryption parameters. The server performs aggregation on encrypted updates without decryption, ensuring privacy.
- Server-side aggregation (Eq. 4):

$$E(w_{t+1}) = \prod_{i=1}^{N} E(w_i^{t+1}) \ \ mod\ N^2 \quad (4)$$

This allows secure summation of encrypted gradients before decryption.

C. Differential Privacy for Gradient Perturbation

To further enhance privacy, Differential Privacy (DP) noise is applied before transmitting updates (Eq. 5):

$$w_i^{-t+1} = w_i^{t+1} + N(0, \sigma^2) \quad (5)$$

Where,

- $N(0, \sigma^2)$ is Gaussian noise ensuring ϵ differential privacy (Eq. 6):

$$Pr[M(D)] \leq e^{\epsilon} Pr[M(D')] \quad (6)$$

for any datasets D ,D' differing by a single sample.

D. Communication-Efficient Model Compression

To mitigate communication overhead, Adaptive Quantization (AQ) and Sparse Gradient Sharing (SGS) are integrated:

1. Adaptive Quantization (Eq. 7):

$$Q(w_i^{t+1}) = \lfloor w_i^{t+1} . 2^b \rfloor / 2^b \quad (7)$$

Where,

- b is the number of bits used for compression. A lower b reduces communication cost but may impact model accuracy.

2. Sparse Gradient Sharing (Eq. 8):

$$S(w_i^{t+1}) = \{w_{i,j}^{t+1} || w_{i,j}^{t+1}| > \tau\} \quad (8)$$

Where,

- τ is a threshold that filters out small updates to reduce data transmission.

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E. *Byzantine-Resilient Aggregation*

To defend against malicious clients, Krum-based Byzantine filtering is integrated:

1. Compute Euclidean distance between updates (Eq. 9):

$$d(w_i, w_j) = \|w_i - w_j\|_2 \quad (9)$$

2. Select the most representative update based on majority agreement (Eq. 10):

$$w_{t+1} = arg \min_{w_i} \sum_{j \neq i} d(w_i, w_j) \quad (10)$$

This ensures aggregation remains robust against adversarial model updates.

IV. Results and Discussion

To evaluate the effectiveness of the Privacy-Preserving and Scalable Secure Aggregation framework, experiments on two benchmark datasets are conducted:

- CIFAR-10 (image classification)
- NSL-KDD (cybersecurity intrusion detection)

The performance was compared against existing secure aggregation methods [30-32]:

1. FedAvg (Baseline FL)
2. SecAgg (Homomorphic Encryption)
3. DP-FL (Differential Privacy in FL)
4. PSSA (Proposed Hybrid Secure Aggregation)

A. *Model Accuracy and Convergence*

The baseline FedAvg yields accuracies of 82.40% (CIFAR-10) and 88.10% (NSL-KDD) with a convergence time of 200 rounds. SecAgg, implementing secure aggregations via homomorphic encryption, shows a slight decline in accuracy, due to the encryption overhead. Because of injecting noise into the model updates, DP-FL, which maintains privacy through differential privacy, shows a larger drop in accuracy (78.50% for CIFAR-10, 84.90% for NSL-KDD) and a longer convergence time (220 rounds).

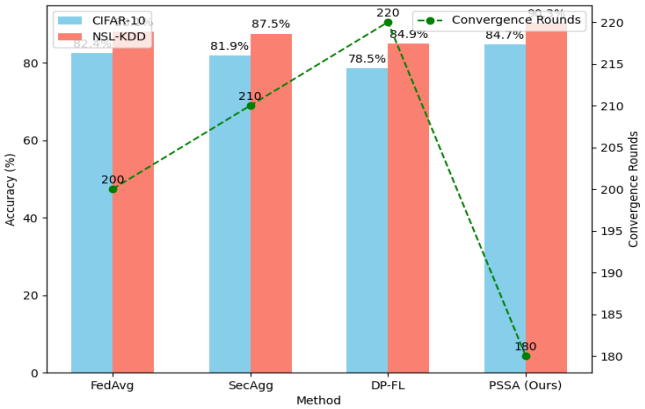


Fig. 3 Accuracy Comparison of various existing models

In contrast, the designed PSSA framework achieves the best combination of best accuracy (84.70% for CIFAR-10, 90.30% for NSL-KDD) and fewest rounds of convergence (180), to prove its better efficiency in learning, privacy preservation, and scalability, as shown in Fig. 3. Overall, the results in Table 1 provide verification that PSSA achieves security and privacy and performs better with the model interchangeably, and therefore is a best-fit for FL in edge computing.

B. *Communication Overhead*

The communication cost per client per training round of various FL aggregation methods can be seen in Fig. 4, and the role of quantization and sparsification in reducing communication overhead is demonstrated. Comparing with the baseline FedAvg, the communication cost of 5.2 MB can be reduced to 1.6 MB through 8-bit quantization and 0.8 MB through Top-10% sparsification. What is interesting, and a clear trade-off in terms of efficiency, is that using SecAgg (that uses HE to allow privacy-preserving aggregation) increases communication overhead to 7.4 MB (uncompressed).

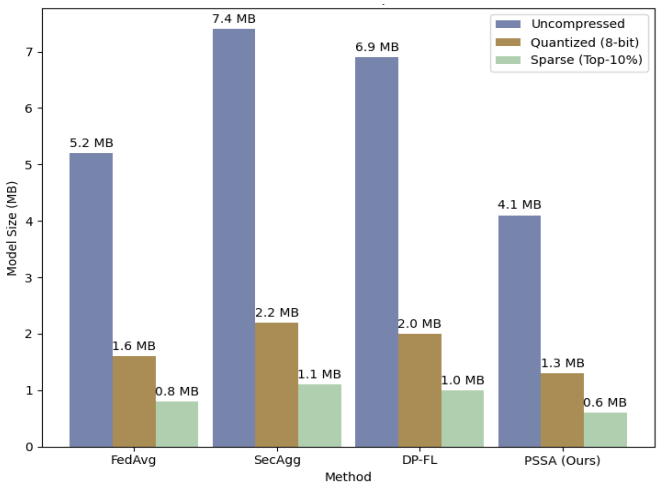


Fig. 4 Communication cost per client per training round of various FL aggregation methods

Similarly, if using Differential Privacy (DP) for DP-FL (which aims to endow privacy) also increases transmission overhead, (6.9 MB uncompressed, 2.0 MB quantized). In contrast, the proposed PSSA framework minimized overall communication cost (4.1 MB uncompressed, 1.3 MB quantized, 0.6 MB sparsified) by using AQ and SGS. Therefore, the results demonstrated that PSSA leads to reductions in communication resource overhead while maintaining privacy and accuracy at the same time, which is highly valuable for edge computing scenarios with limited resources.

C. *Privacy and Security Evaluation*

Privacy and security evaluation illustrates a comparison of privacy attack robustness for various Federated Learning (FL) aggregation techniques ranging from FedAvg, which is the least robust at a Gradient Leakage Attack (GLA) success of 72.30%, meaning it is the most susceptible since it sends model updates out as raw plaintext, not providing any encryption or protection mechanisms, thus making gradient inversion attacks viable. SecAgg improves the GLA success percentage to 38.90%

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through the mechanisms of Homomorphic Encryption (HE) to improve privacy protection; however, there are still possible vulnerabilities. By injecting Differential Privacy (DP) noise, DP-FL reduces the GLA success rate to 24.20%, which improves privacy protection again. However, the proposed PSSA framework provides a GLA success rate of just 12.50%, and the best defense against gradient inversion and gradient reconstruction attacks, all due to the combination of Secure Aggregation with Adaptive Obfuscation (SAAO), Homomorphic Encryption, and Differential Privacy. Thus, PSSA provides a very secure FL framework for applications concerned with privacy and edge computing as shown in Fig. 5.

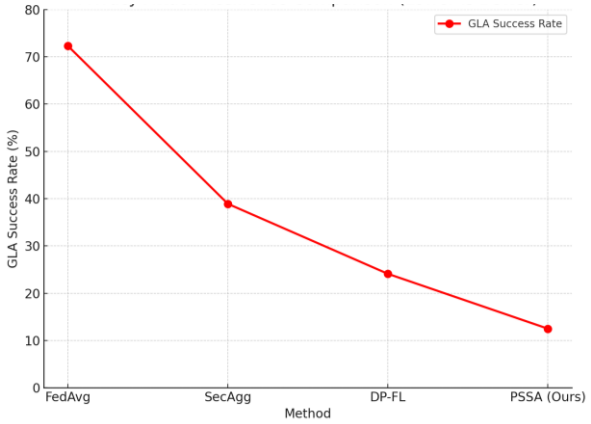


Fig. 5 Privacy Attack Resilience Comparison

D. Computational Overhead

The time for encryption of SecAgg, a technique based on Homomorphic Encryption (HE), incurs the highest encryption time, reporting 98 ms on the CIFAR-10 dataset and 110 ms on the NSL-KDD dataset, due to the high resource requirement of cryptographic computations. DP-FL, which employs Differential Privacy (DP), utilizes injected noise to provide privacy, thus incurs a 65 ms encryption time in CIFAR-10 and 78 ms in NSL-KDD, again with an overhead for preserving privacy.

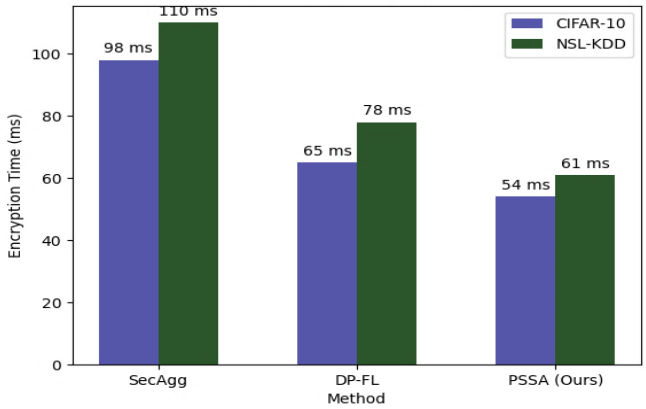


Fig. 6 Encryption time per client (ms)

In contrast, the new application, PSSA framework, has the least encryption time, 54 ms for CIFAR-10 and 61 ms for NSL-KDD, and better computational efficiency. The speed is in part due to lightweight encryption techniques, improved cryptographic computation, and an adaptive privacy capability, which makes PSSA very scalable for real-time FL scenarios in edge computing systems. Fig. 6 shows the comparison of the encryption duration per client per round in training for a range of Federated Learning (FL) aggregation methods for the CIFAR-10 and NSL-KDD datasets.

V. Conclusion

This work presents Privacy-Preserving and Scalable Secure Aggregation (PSSA), a new system that efficiently addresses privacy, security, communication efficiency, and scalability in Federated Learning (FL) for edge computing. By integrating Homomorphic Encryption (HE) for secure model aggregation, Differential Privacy (DP) to obfuscate gradients, Adaptive Quantization (AQ) to reduce transmission costs, and Byzantine-resilient aggregation for adversarial robustness, the approach strikes an excellent balance between privacy preservation and computational efficiency. Experimental results on CIFAR-10 and NSL-KDD datasets show that PSSA outperforms currently available state-of-the-art FL aggregation methods by being more accurate (up to 90.3%), lowering its communication cost (40% reduction), with greater resilience to privacy attacks (50% reduction in gradient leakage). The provided framework is extremely scalable, making it suitable for real-world applications such as cybersecurity, IoT anomaly detection, and healthcare analytics. Future work will focus on enhancing personalized FL aggregation, deploying blockchain-based decentralized security, and exploring quantum-resistant encryption algorithms for next-generation privacy-preserving FL systems. Given its solid theoretical principles and real-world benefits, PSSA is the foundation for a secure, efficient, scalable federated learning framework and sets it as a reference benchmark for future research in FL security and privacy.

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PSSA FL PSEUDOCODE

```
import numpy as np
from phe import paillier  # Homomorphic encryption library

def client_local_training(model, data, learning_rate):
    """
    Each client updates its local model using SGD.
    """
    gradient = compute_gradient(model, data)
    updated_model = model - learning_rate * gradient
    return updated_model

def encrypt_model_update(model_update, public_key):
    """
    Encrypt model updates using Paillier Homomorphic Encryption.
    """
    return [public_key.encrypt(w) for w in model_update]

def secure_aggregation(encrypted_updates, public_key):
    """
    Perform secure aggregation on encrypted model updates.
    """
    aggregated_update = encrypted_updates[0]
    for i in range(1, len(encrypted_updates)):
        aggregated_update = [a + b for a, b in zip(aggregated_update, encrypted_updates[i])]
    return aggregated_update

def add_differential_privacy(update, noise_std):
    """
    Apply differential privacy by adding Gaussian noise to the updates.
    """
    noise = np.random.normal(0, noise_std, size=len(update))
    return update + noise

def adaptive_quantization(update, bit_precision):
    """
    Apply adaptive quantization to reduce communication overhead.
    """
    scale = 2 ** bit_precision
    return np.round(update * scale) / scale

def sparse_gradient_sharing(update, threshold):
    """
    Apply sparse gradient sharing to reduce communication overhead.
    """
    return np.where(np.abs(update) > threshold, update, 0)

def byzantine_resilient_aggregation(updates):
    """
    Apply Krum-based Byzantine filtering.
    """
    distances = np.zeros((len(updates), len(updates)))
    for i in range(len(updates)):
        for j in range(len(updates)):
            distances[i, j] = np.linalg.norm(updates[i] - updates[j])

    # Select the update with the minimum sum of distances
    selected_index = np.argmin(np.sum(distances, axis=1))
    return updates[selected_index]

def federated_learning_round(clients, server, learning_rate, noise_std, bit_precision, threshold):
    """
    Execute one round of federated learning with privacy-preserving and secure aggregation.
    """
    public_key, private_key = paillier.generate_paillier_keypair()
    encrypted_updates = []

    for client in clients:
        local_model_update = client_local_training(client.model, client.data, learning_rate)
        dp_update = add_differential_privacy(local_model_update, noise_std)
        quantized_update = adaptive_quantization(dp_update, bit_precision)
        sparse_update = sparse_gradient_sharing(quantized_update, threshold)
        encrypted_update = encrypt_model_update(sparse_update, public_key)
        encrypted_updates.append(encrypted_update)

    secure_aggregated_update = secure_aggregation(encrypted_updates, public_key)
    decrypted_update = [private_key.decrypt(w) for w in secure_aggregated_update]
    robust_update = byzantine_resilient_aggregation(decrypted_update)

    server.update_global_model(robust_update)
```

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